

Divided Difference

General formulae

$$\begin{array}{l}
 x_0 \rightarrow f(x_0) \\
 x_1 \rightarrow f(x_1) \\
 x_2 \rightarrow f(x_2) \\
 x_3 \rightarrow f(x_3)
 \end{array}
 \begin{array}{l}
 \nearrow f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{x_1 - x_0} \\
 \nearrow f(x_1, x_2) = \frac{f(x_2) - f(x_1)}{x_2 - x_1} \\
 \nearrow f(x_2, x_3) = \frac{f(x_3) - f(x_2)}{x_3 - x_2}
 \end{array}
 \begin{array}{l}
 \nearrow f(x_0, x_1, x_2) = \frac{f(x_1, x_2) - f(x_0, x_1)}{x_2 - x_0} \\
 \nearrow f(x_1, x_2, x_3) = \frac{f(x_2, x_3) - f(x_1, x_2)}{x_3 - x_1}
 \end{array}
 \nearrow f(x_0, x_1, x_2, x_3) = \frac{f(x_1, x_2, x_3) - f(x_0, x_1, x_2)}{x_3 - x_0}$$

⇒ Polynomials "underlined"

$$P_n(x) = \underbrace{f(x_0)}_{\text{deg } 0} + \underbrace{f(x_0, x_1)}_{\text{deg } 1} (x - x_0) + \underbrace{f(x_0, x_1, x_2)}_{\text{deg } 2} (x - x_0)(x - x_1) + \underbrace{f(x_0, x_1, x_2, x_3)}_{\text{deg } 3} (x - x_0)(x - x_1)(x - x_2) + \dots$$

Q2

2. Use Eq. (3.10) or Algorithm 3.2 to construct interpolating polynomials of degree one, two, and three for the following data. Approximate the specified value using each of the polynomials.

a. $f(0.43)$ if $f(0) = 1$, $f(0.25) = 1.64872$, $f(0.5) = 2.71828$, $f(0.75) = 4.48169$

$x \Rightarrow 0.43$

$$\begin{array}{l}
 x_0 = 0 \\
 x_1 = 0.25 \\
 x_2 = 0.5 \\
 x_3 = 0.75
 \end{array}
 \begin{array}{l}
 \Rightarrow f(x_0, x_1) = 2.59488 \\
 \Rightarrow f(x_1, x_2) = 4.27824 \\
 \Rightarrow f(x_2, x_3) = 7.05364
 \end{array}
 \begin{array}{l}
 \Rightarrow f(x_0, x_1, x_2) = 3.66672 \\
 \Rightarrow f(x_1, x_2, x_3) = 5.5508
 \end{array}
 \Rightarrow f(x_0, x_1, x_2, x_3) = 2.5121066$$

$$P_1(0.43) = f(x_0) + f(x_0, x_1)(x - x_0) = 2.1157984$$

$$P_2(0.43) = P_1(x) + f(x_0, x_1, x_2)(x - x_0)(x - x_1) = 2.3996025$$

$$P_3(0.43) = P_2(x) + f(x_0, x_1, x_2, x_3)(x - x_0)(x - x_1)(x - x_2) = 2.3859919$$

Newton's forward method &

$$\begin{array}{l}
 x_0 \rightarrow f(x_0) \\
 x_1 \rightarrow f(x_1) \\
 x_2 \rightarrow f(x_2) \\
 x_3 \rightarrow f(x_3)
 \end{array}
 \begin{array}{l}
 \text{deg 1} \\
 \begin{array}{l}
 f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{h} \\
 f(x_1, x_2) = \frac{f(x_2) - f(x_1)}{h} \\
 f(x_2, x_3) = \frac{f(x_3) - f(x_2)}{h}
 \end{array} \\
 \text{deg 2} \\
 \begin{array}{l}
 f(x_0, x_1, x_2) = \frac{f(x_0, x_1) - f(x_1, x_2)}{2h} \\
 f(x_1, x_2, x_3) = \frac{f(x_1, x_2) - f(x_2, x_3)}{2h}
 \end{array} \\
 \text{deg 3} \\
 f(x_0, x_1, x_2, x_3) = \frac{f(x_0, x_1, x_2) - f(x_1, x_2, x_3)}{3h}
 \end{array}$$

$$\Rightarrow h = x_1 - x_0 = x_2 - x_1 = x_3 - x_2$$

$$\Rightarrow s = \frac{x - x_0}{h}, \text{ in forward method "s" is +ve}$$

$\Rightarrow$ polynomial &

$$P_n(x) = f(x_0) + f(x_0, x_1)sh + f(x_0, x_1, x_2)s(s-1)h^2 + f(x_0, x_1, x_2, x_3)s(s-2)h^3 + \dots$$

Q40

4. Use the Newton forward-difference formula to construct interpolating polynomials of degree one, two, and three for the following data. Approximate the specified value using each of the polynomials.

$$\begin{array}{l}
 x_0 \quad x_1 \quad x_2 \\
 f(0.18) \text{ if } f(0.1) = -0.29004986, f(0.2) = -0.56079734, f(0.3) = -0.81401972, \\
 x_3 \quad f(0.4) = -1.0526302
 \end{array}$$

$$x \Rightarrow 0.18$$

$$\begin{array}{l}
 x_0 = 0.1 \\
 x_1 = 0.2 \\
 x_2 = 0.3 \\
 x_3 = 0.4
 \end{array}
 \begin{array}{l}
 > f(x_0, x_1) = -0.3384343 \\
 > f(x_1, x_2) = -0.3165279 \\
 > f(x_2, x_3) = -0.2982631
 \end{array}
 \begin{array}{l}
 > f(x_0, x_1, x_2) = 0.0136915 \\
 > f(x_1, x_2, x_3) = 0.0114155
 \end{array}
 > f(x_0, x_1, x_2, x_3) = -0.0009483$$

$$h = 0.1$$

$$s = \frac{0.18 - 0.1}{0.1} = 0.8$$

$$P(0.18) = f(x_0) + f(x_0, x_1)s \cdot h + f(x_0, x_1, x_2)s(s-1)h^2 + f(x_0, x_1, x_2, x_3)s(s-1)h^3$$

$$= -0.3171464$$

Q9%

9. a. Approximate $f(0.05)$ using the following data and the Newton forward-difference formula:

	x_0	x_1	x_2	x_3	x_4
x	0.0	0.2	0.4	0.6	0.8
$f(x)$	1.00000	1.22140	1.49182	1.82212	2.22554

$$x \Rightarrow 0.05$$

$$\begin{array}{l}
 x_0 = 0 \\
 x_1 = 0.2 \\
 x_2 = 0.4 \\
 x_3 = 0.6 \\
 x_4 = 0.8
 \end{array}
 \begin{array}{l}
 \left. \begin{array}{l} \\ \\ \\ \end{array} \right\} \begin{array}{l} f(x_0, x_1) = 1.107 \\ f(x_1, x_2) = 1.3521 \\ f(x_2, x_3) = 1.6515 \\ f(x_3, x_4) = 2.0171 \end{array} \\
 \left. \begin{array}{l} \\ \\ \end{array} \right\} \begin{array}{l} f(x_0, x_1, x_2) = 0.61275 \\ f(x_1, x_2, x_3) = 0.7485 \\ f(x_2, x_3, x_4) = 0.914 \end{array} \\
 \left. \begin{array}{l} \\ \end{array} \right\} \begin{array}{l} f(x_0, x_1, x_2, x_3) = 0.22625 \\ f(x_1, x_2, x_3, x_4) = 0.275833333 \end{array}
 \end{array}
 \begin{array}{l}
 \xrightarrow{\quad} \\
 \xrightarrow{\quad} \\
 \xrightarrow{\quad} \\
 \xrightarrow{\quad}
 \end{array}
 \begin{array}{l}
 f(x_0, x_1, x_2, x_3, x_4) = 0.04010416625
 \end{array}$$

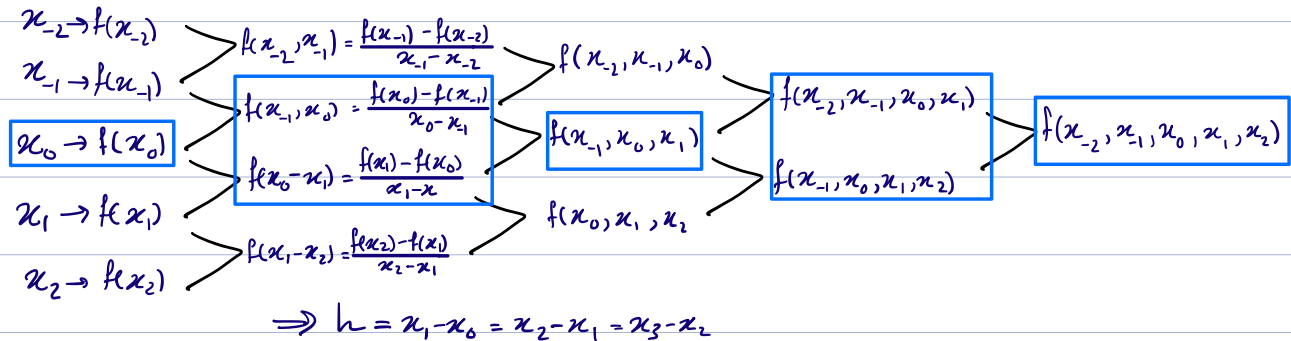
$$h = 0.2$$

$$s = \frac{0.5 - 0}{0.2} = 2.5$$

$$p(0.05) = f(x_0) + sh(f(x_0, x_1)) + s(s-1)h^2(f(x_0, x_1, x_2)) + s(s-1)h^3(f(x_0, x_1, x_2, x_3)) + s(s-1)h^4(f(x_0, x_1, x_2, x_3, x_4)) =$$

$$= 1.6524406$$

Centered difference (Stirling formula) 8



$\Rightarrow$ polynomial 8

$$P_n(x) = f(x_0) + \frac{Sh}{2} [f(x_{-1}, x_0) + f(x_0, x_1)] + \frac{S^2 h^2}{2} [f(x_{-2}, x_{-1}, x_0, x_1) + \frac{S(S^2-1)}{2} \cdot h^2 [f(x_{-2}, x_{-1}, x_0, x_1) + f(x_{-1}, x_0, x_1, x_2)]] + S^2(S^2-1)h^4 [f(x_{-2}, x_{-1}, x_0, x_1, x_2)] + \dots$$

above $x_0 \Rightarrow x_{-1}, x_{-2}, \dots$

below $x_0 \Rightarrow x_1, x_2, \dots$

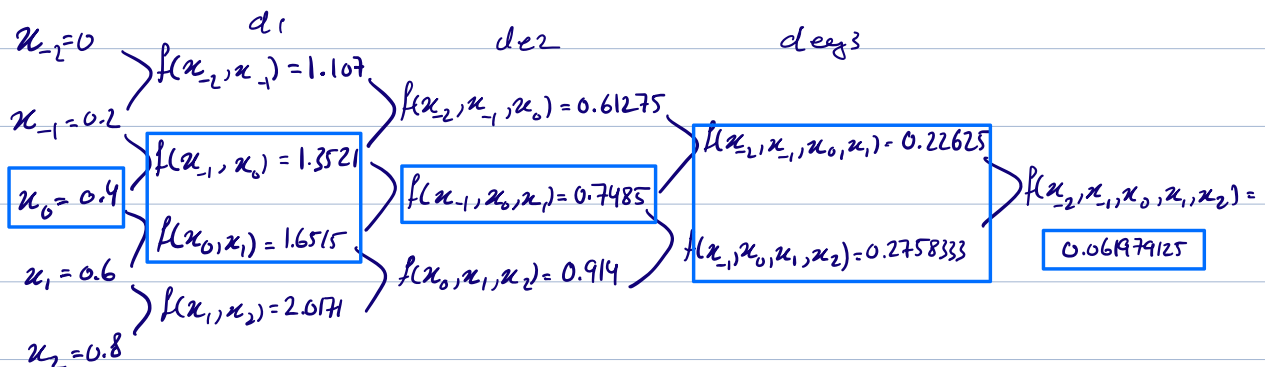
Q90

c. Use Stirling's formula to approximate $f(0.43)$.

x	0.0	0.2	0.4	0.6	0.8
f(x)	1.00000	1.22140	1.49182	1.82212	2.22554

$$h = 0.2$$

$$S = \frac{x - x_0}{h} = \frac{0.43 - 0.4}{0.2} = .15$$



$$p(0.43) = \overset{\text{deg 1}}{f(x_0)} + \frac{sh}{2} (\overset{\text{deg 1}}{f(x_{-1}, x_0)} + \overset{\text{deg 1}}{f(x_0, x_1)}) + \overset{\text{deg 2}}{s^2 h^2 f(x_{-1}, x_0, x_1)} + \frac{s(s^2-1)h^3}{2} (\overset{\text{deg 3}}{f(x_{-1}, x_0, x_1)}) +$$

$$s^2(s^2-1)h^4 (f(x_{-2}, x_{-1}, x_0, x_1, x_2)) = 1.53722913 \rightarrow \text{not scene}$$

Newton's Backward method

$$\begin{array}{l}
 x_0 \rightarrow f(x_0) \\
 x_1 \rightarrow f(x_1) \\
 x_2 \rightarrow f(x_2) \\
 x_3 \rightarrow f(x_3)
 \end{array}
 \begin{array}{l}
 \text{deg 1} \\
 f(x_0, x_1) = \frac{f(x_1) - f(x_0)}{h} \\
 f(x_1, x_2) = \frac{f(x_2) - f(x_1)}{h} \\
 f(x_2, x_3) = \frac{f(x_3) - f(x_2)}{h}
 \end{array}
 \begin{array}{l}
 \text{deg 2} \\
 f(x_0, x_1, x_2) = \frac{f(x_0, x_1) - f(x_1, x_2)}{2h} \\
 f(x_1, x_2, x_3) = \frac{f(x_1, x_2) - f(x_2, x_3)}{2h}
 \end{array}
 \begin{array}{l}
 \text{deg 3} \\
 f(x_0, x_1, x_2, x_3) = \frac{f(x_0, x_1, x_2) - f(x_1, x_2, x_3)}{3h}
 \end{array}$$

$\Rightarrow$ polynomial

$$\begin{aligned}
 P_n(x) = & f(x_0) + \frac{s \cdot h}{2} [f(x_{-1}, x_0) + f(x_0, x_1)] + \frac{s^2 h^2}{2} [f(x_{-1}, x_0, x_1)] + \frac{s(s^2 - 1)}{2} \cdot h^3 [f(x_{-2}, x_{-1}, x_0, x_1) + f(x_{-1}, x_0, x_1, x_2)] \\
 & + \frac{s^2(s^2 - 1)h^4}{2} [f(x_{-2}, x_{-1}, x_0, x_1, x_2)] + \dots
 \end{aligned}$$

above $x_0 \Rightarrow x_{-1}, x_{-2}, \dots$

below $x_0 \Rightarrow x_1, x_2, \dots$

$$\Rightarrow h = x_1 - x_0 = x_2 - x_1 = x_3 - x_2$$

$$\Rightarrow s = \frac{x - x_0}{h}, \text{ In Backward method "s" is -ve}$$

9. a. $f(0.05) \approx 1.02126$

b. $f(0.02) \approx 1.91555$

c. $f(0.43) \approx 1.53725$